Lesson End Project
Implementing Advanced React Routing Features

Project agenda: To develop a React application with advanced routing capabilities using Vite, focusing on nested routes, dynamic URL parameters, custom error handling, and component-level styling using Styled Components. The project will ensure structured navigation, improved user experience, and reliable access management.

Description: You are tasked with building a React application that integrates advanced routing features to optimize navigation and access management. This includes implementing nested routes, dynamic URL parameters, custom error handling with a 404 page, and component-level styling using Styled Components.

Tools required: Visual Studio Code and Node.js

Prerequisites: None

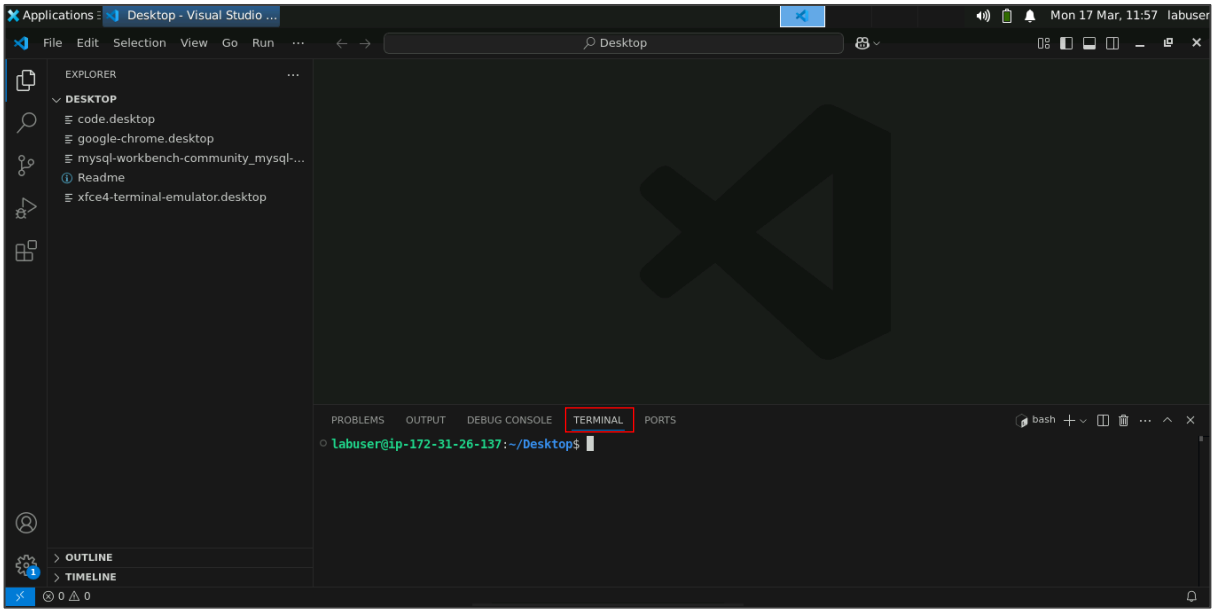
Expected deliverables: A fully functional React-based application built with Vite, incorporating advanced routing capabilities. The application will support structured navigation using nested routes, dynamic route parameters for personalized content, and programmatic navigation for seamless redirection. It will include a custom 404 error page for handling invalid routes and component-level styling using Styled Components to enhance the user experience.

Steps to be followed:

1. Set up a new React project using Vite
2. Structure the application
3. Implement routing features
4. Set up context for user management
5. Set up routing in App.jsx
6. Update main.jsx
7. Run and verify the application

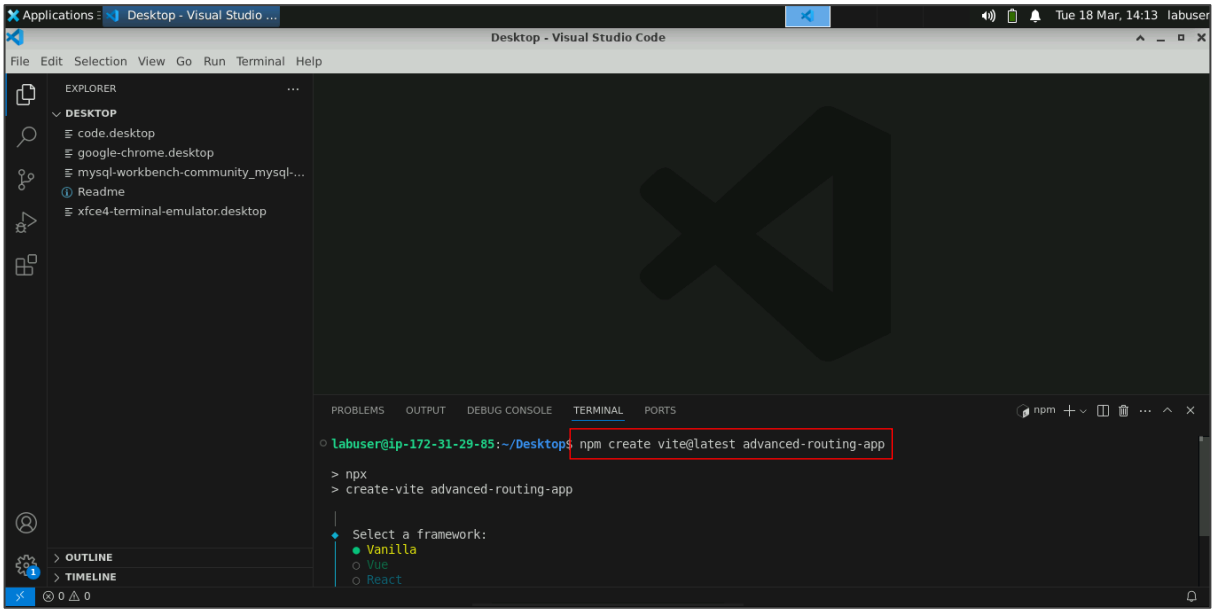
Step 1: Set up a new React project using Vite

1.1 Navigate to VS Code and open the terminal

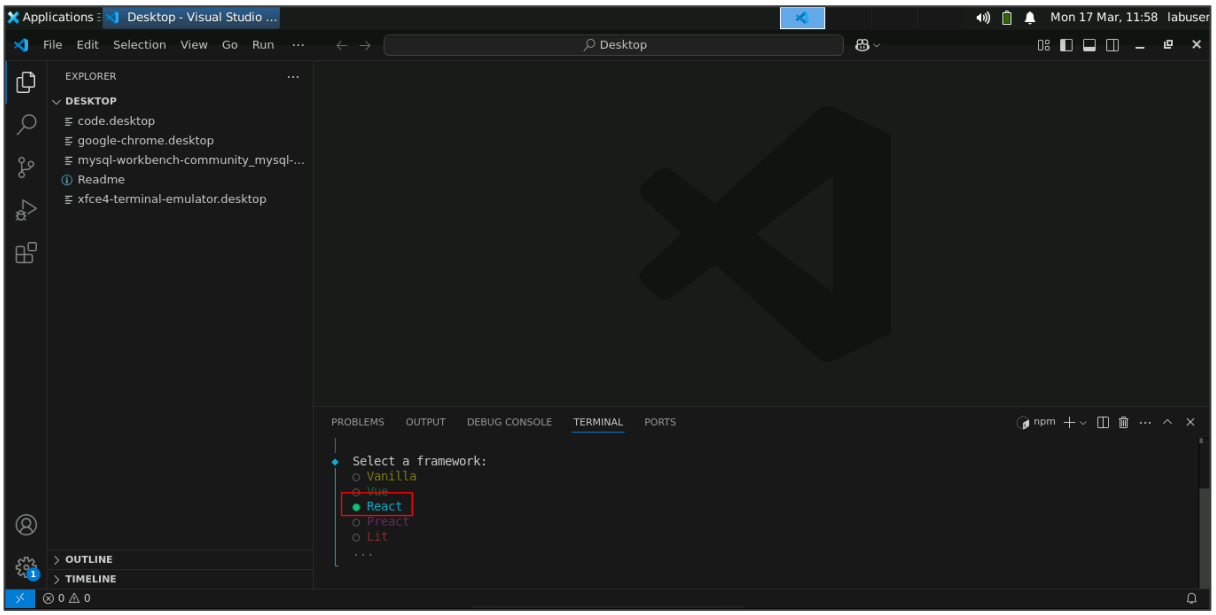


1.2 Run the following command to create a new React project using Vite:

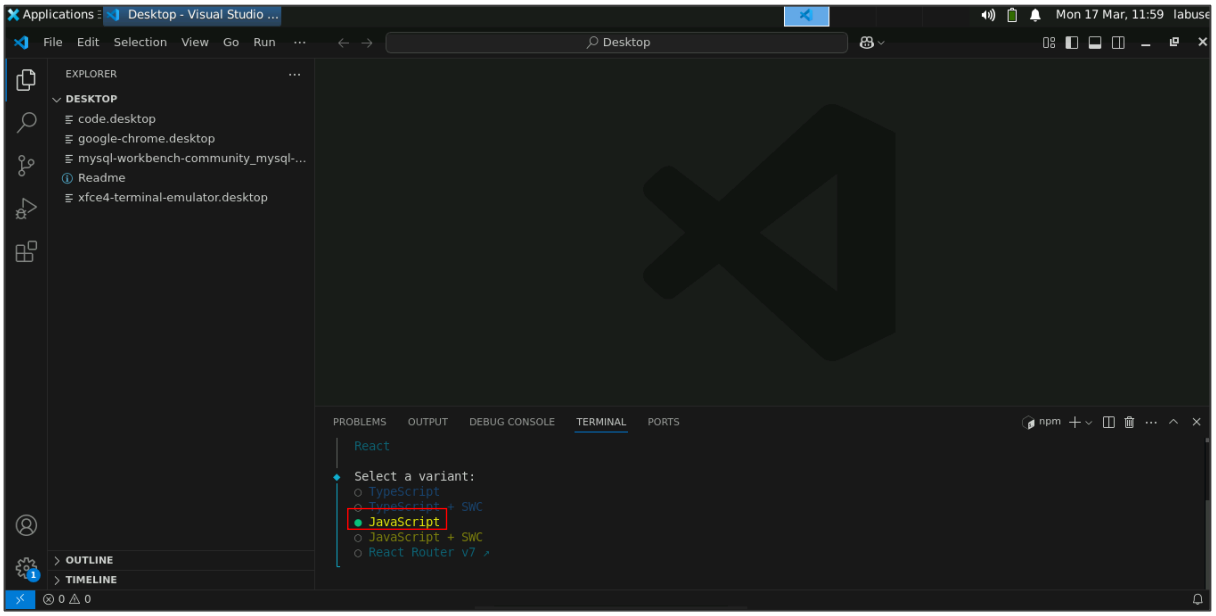
npm create vite@latest advanced-routing-app



1.3 Select **React** and press enter

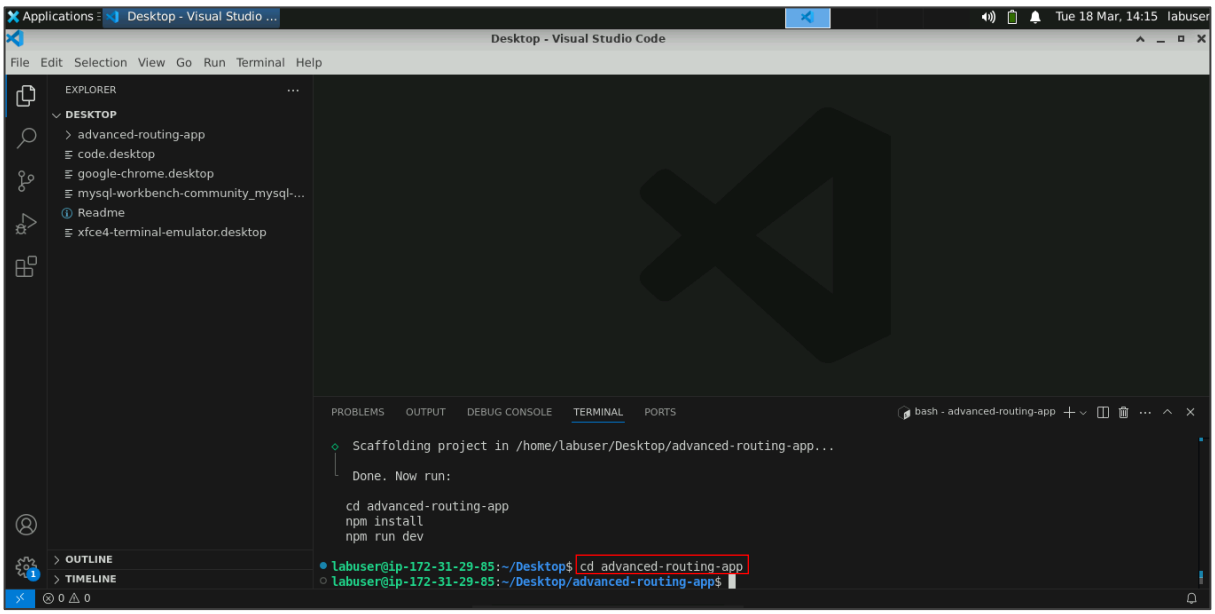


1.4 Select **JavaScript** and press enter



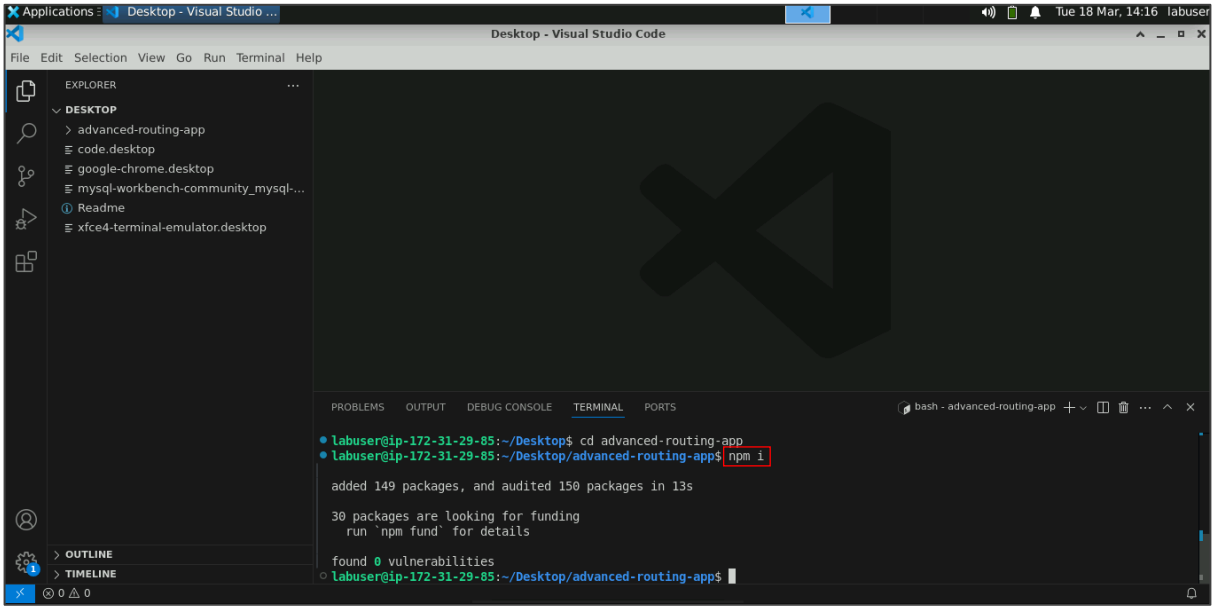
1.5 Run the following command to navigate to the project folder:

cd advanced-routing-app



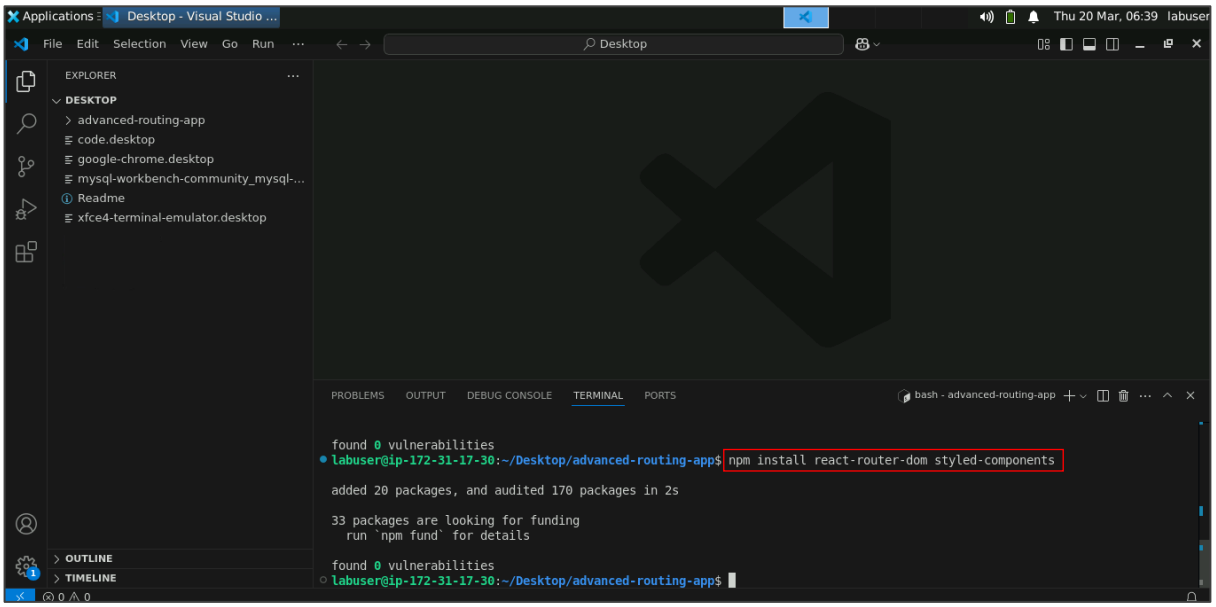
1.6 Run the following command to install dependencies:

npm i



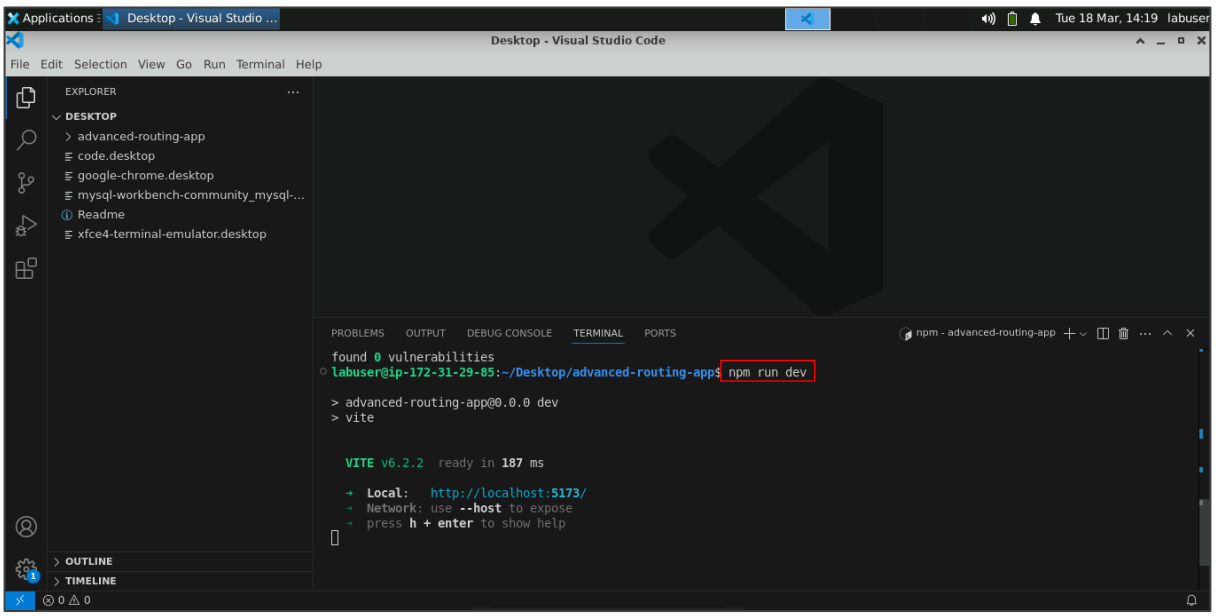
1.7 Run the following command to install React Router and Styled Components:

npm install react-router-dom styled-components

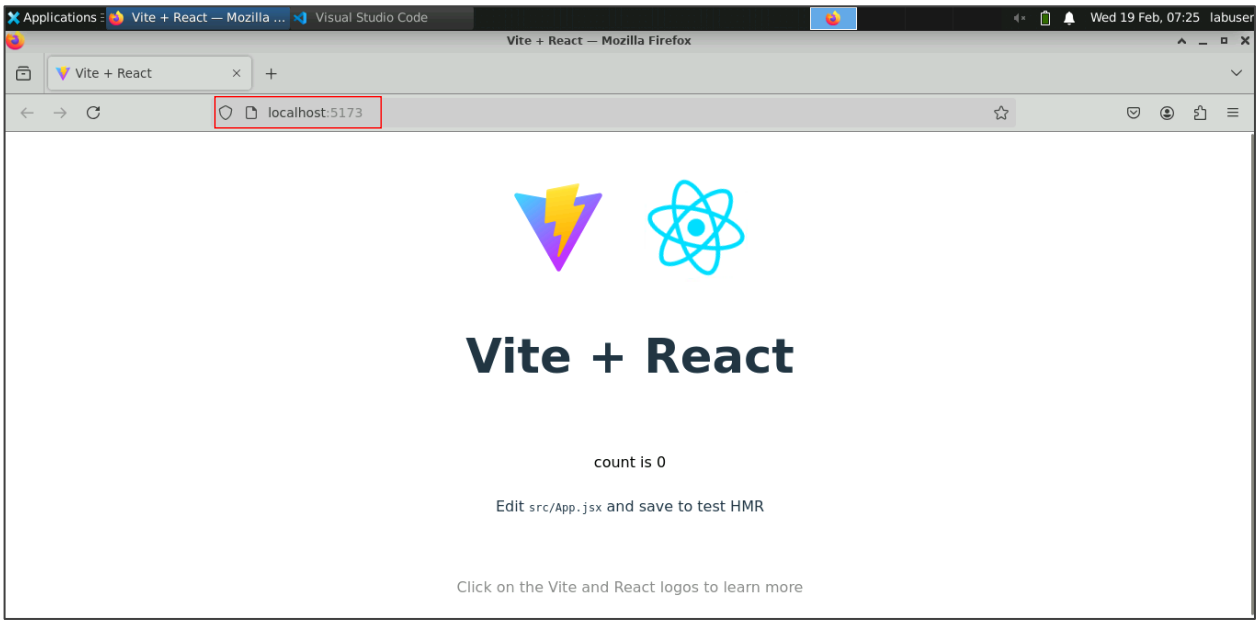


1.8 Run the following command to start the development server:

npm run dev

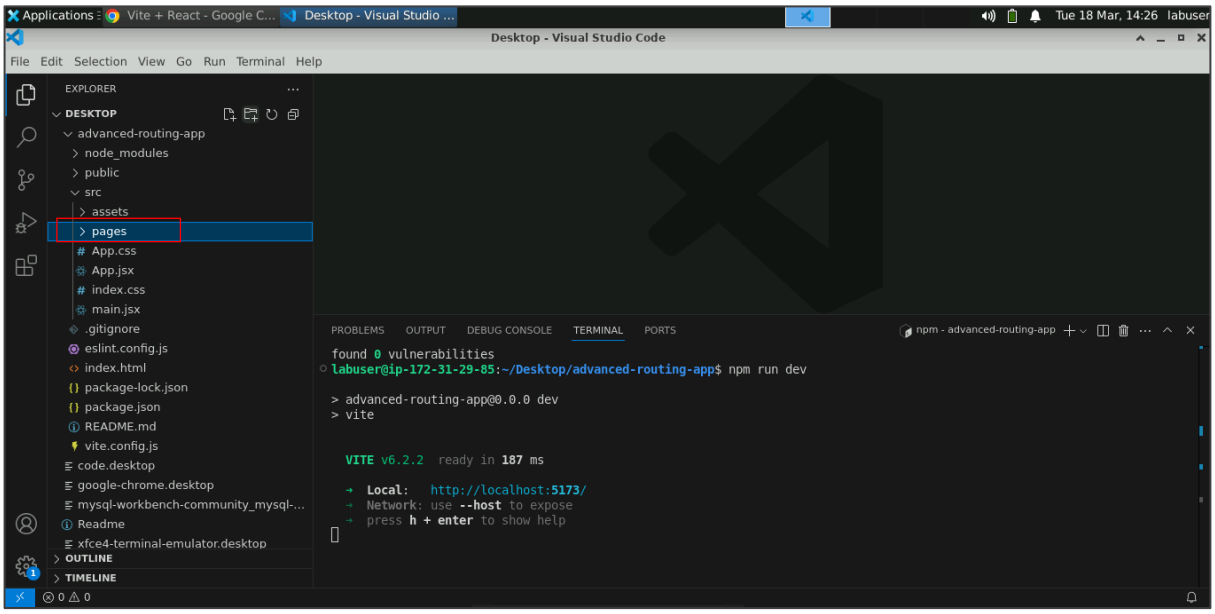


1.9 Navigate to the web browser and paste the link below:
http://localhost:5173

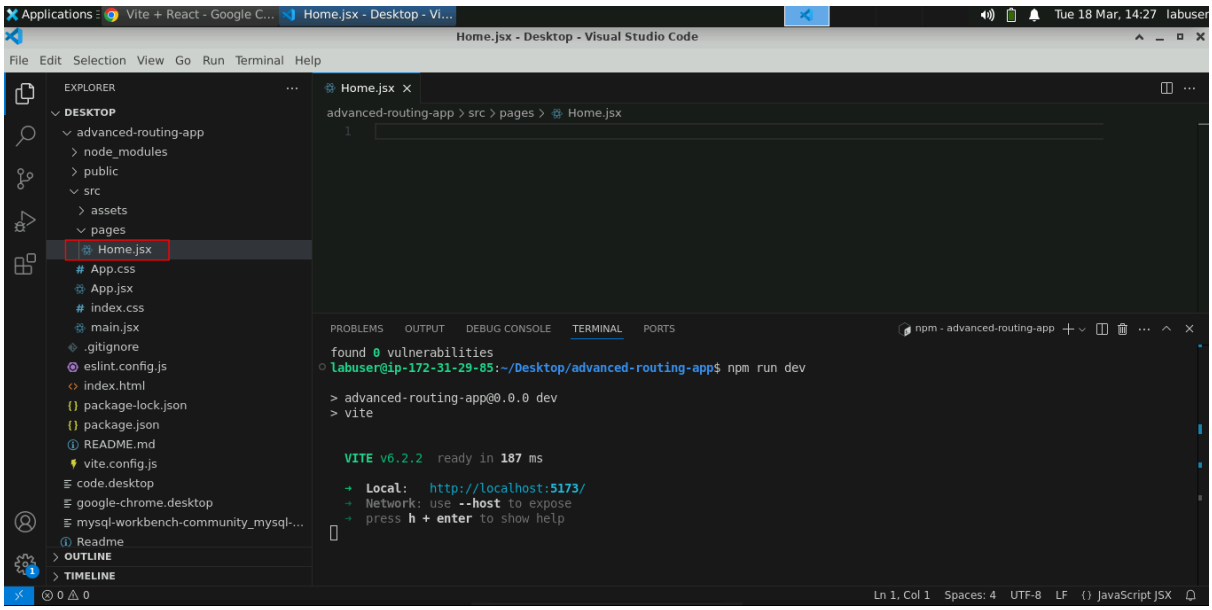


Step 2: Structure the application

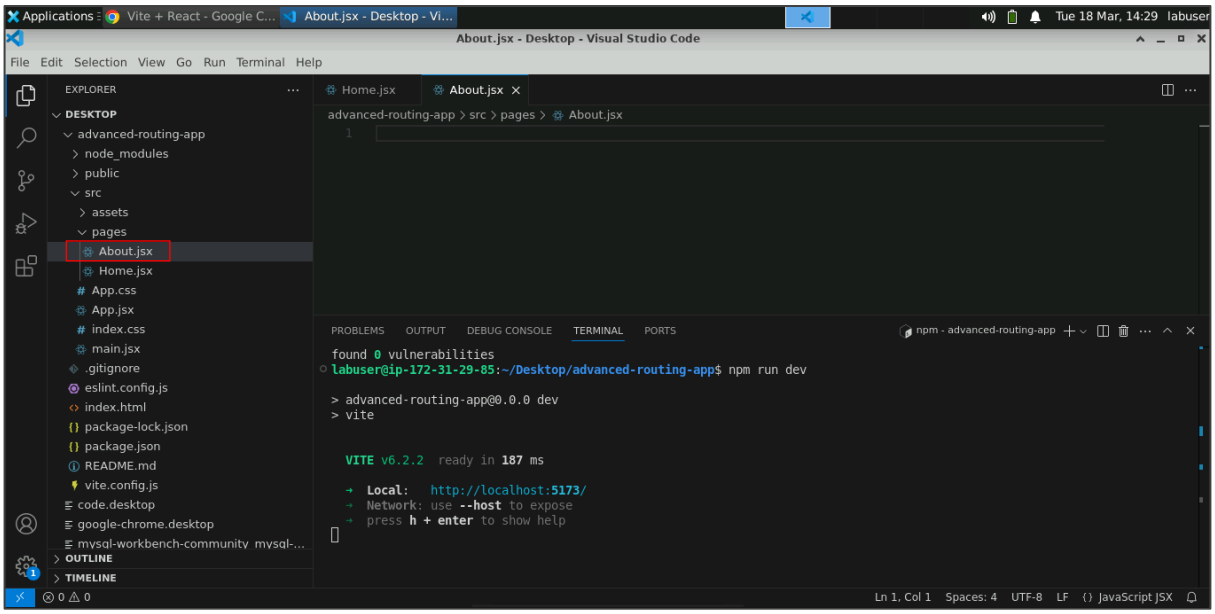
2.1 Inside the **src** folder, create a new folder named **pages**



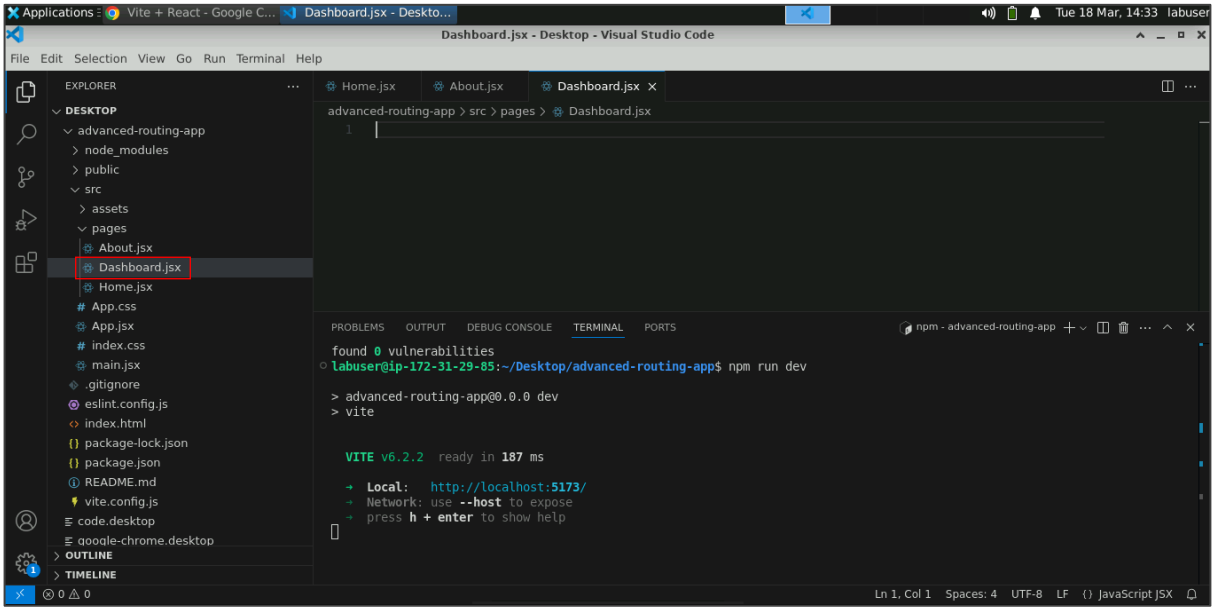
2.2 Inside the **pages** folder, create a new file named **Home.jsx**



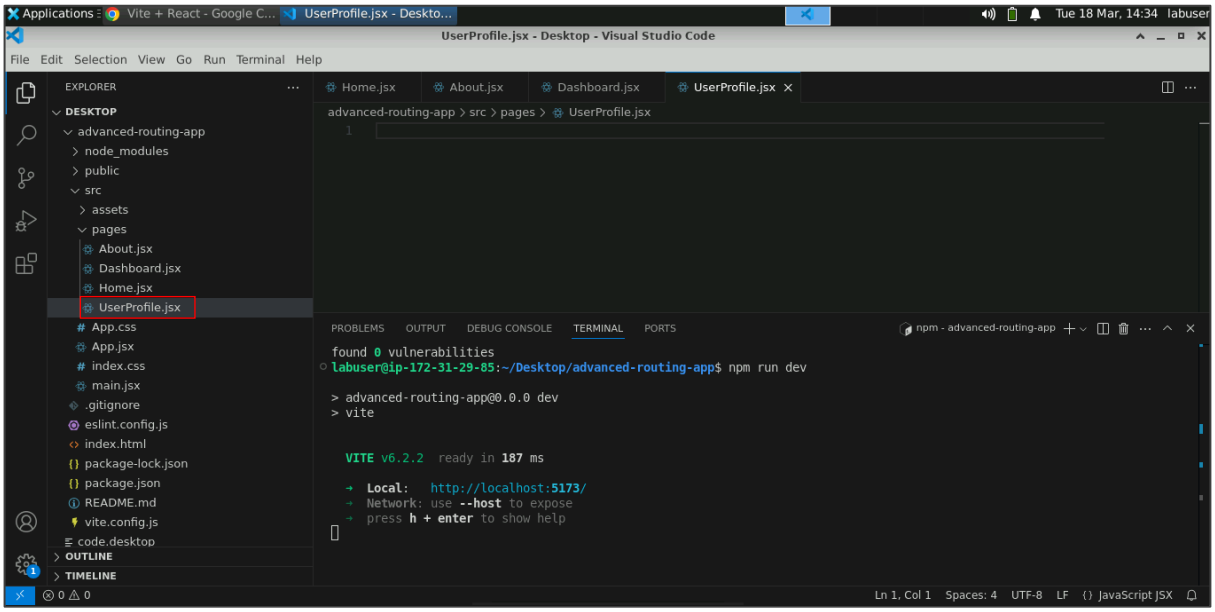
2.3 Inside the **pages** folder, create a new file named **About.jsx**



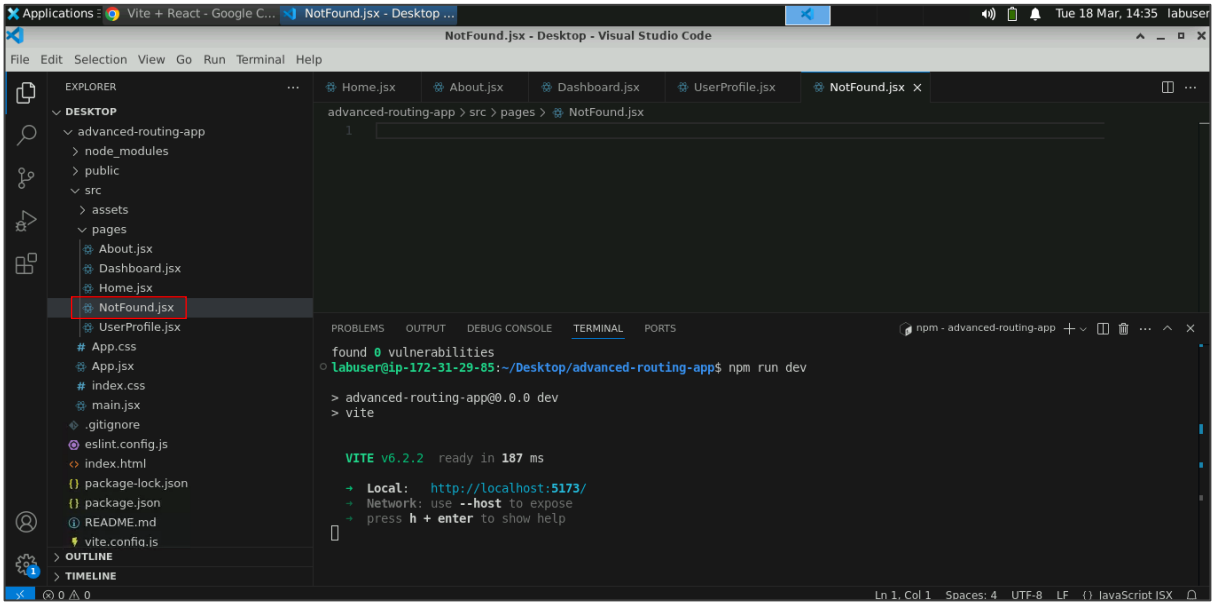
2.4 Inside the **pages** folder, create a new file named **Dashboard.jsx**



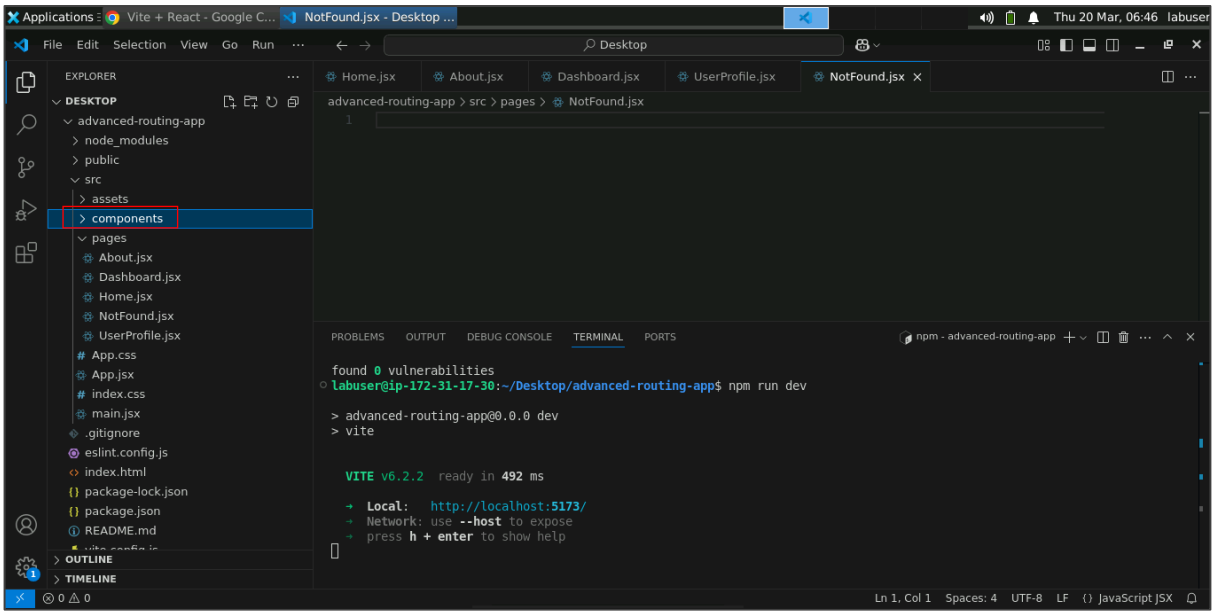
2.5 Inside the **pages** folder, create a new file named **UserProfile.jsx**



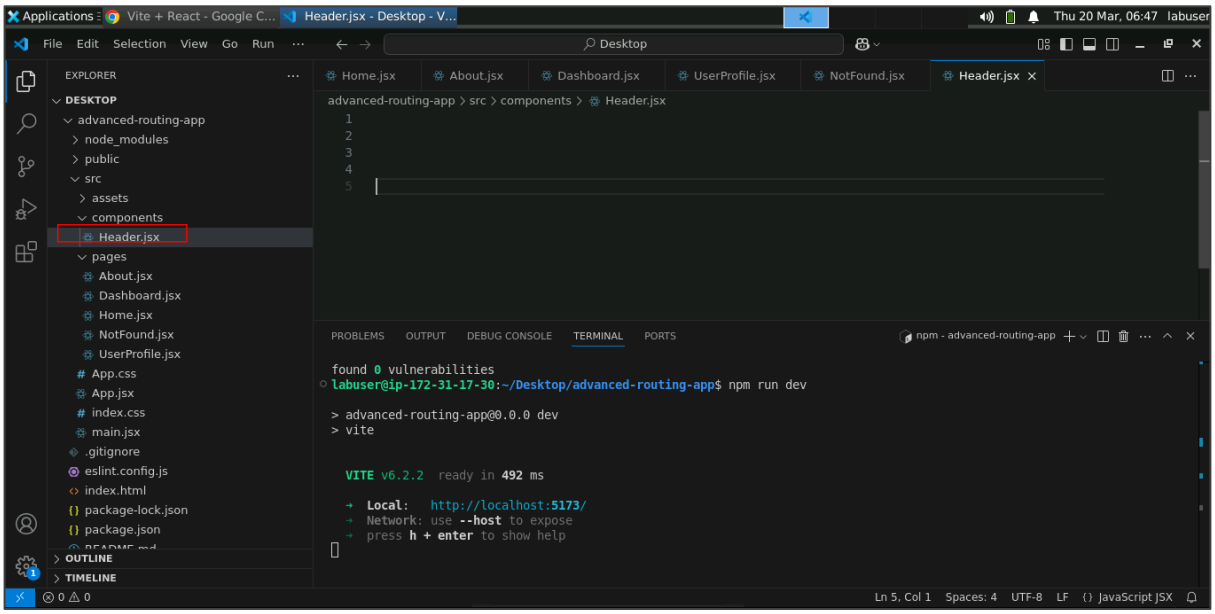
2.6 Inside the **pages** folder, create a new file named **NotFound.jsx**



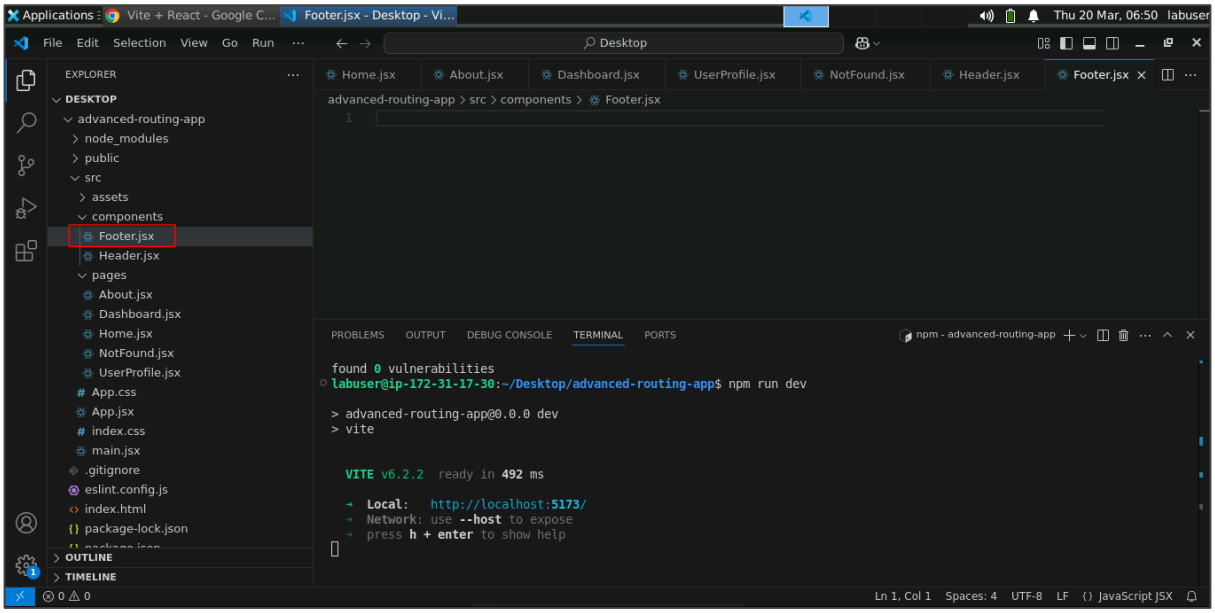
2.7 Inside the **src** folder, create a new folder named **components**



2.8 Inside the **components** folder, create a new file named **Header.jsx**



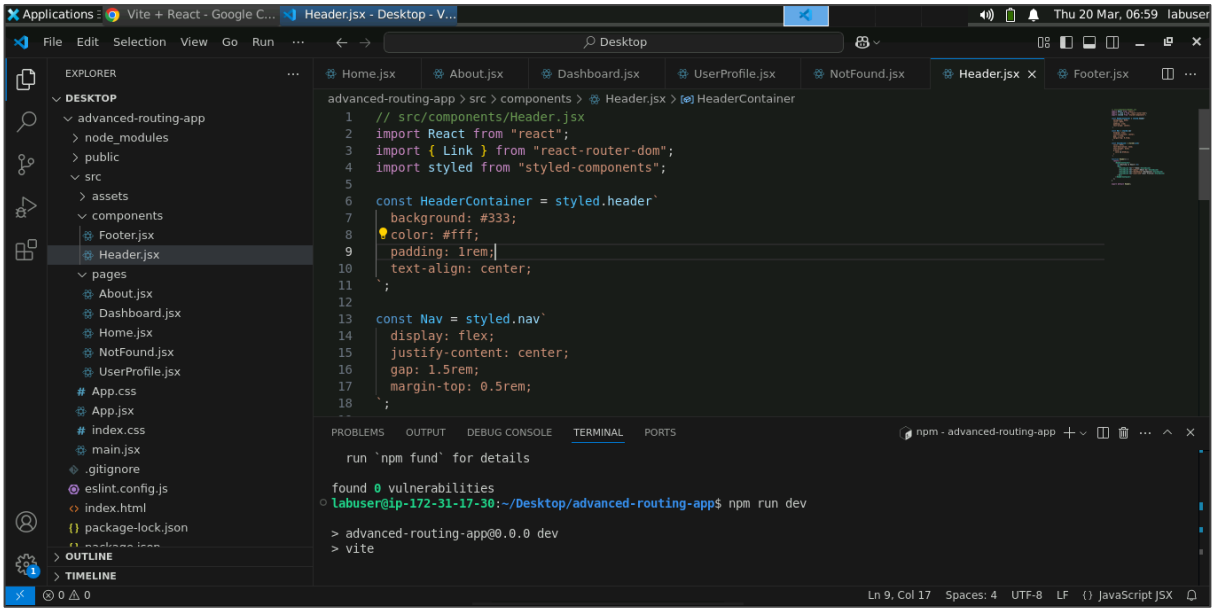
2.9 Inside the **components** folder, create a new file named **Footer.jsx**



Step 3: Implement routing features

3.1 Paste the code below in **Header.jsx**:

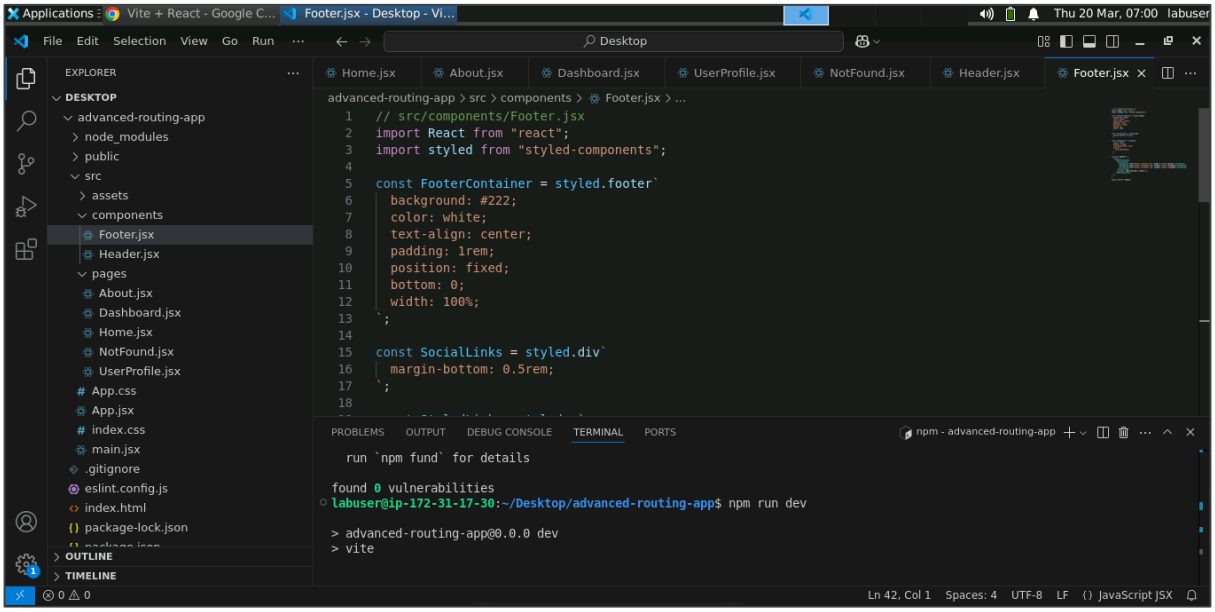
```
// src/components/Header.jsx
import React from "react";
import { Link } from "react-router-dom";
import styled from "styled-components";

const HeaderContainer = styled.header`
  background: #333;
  color: #fff;
  padding: 1rem;
  text-align: center;
`;

const Nav = styled.nav`
  display: flex;
  justify-content: center;
  gap: 1.5rem;
  margin-top: 0.5rem;
`;

const StyledLink = styled(Link)`
  color: #fff;
  text-decoration: none;
  font-weight: bold;
  &:hover {
    color: #f39c12;
  }
`;

function Header() {
  return (
    <HeaderContainer>
      <h1>Routing in React</h1>
      <Nav>
        <StyledLink to="/">Home</StyledLink>
        <StyledLink to="/about">About Us</StyledLink>
        <StyledLink to="/dashboard">Dashboard</StyledLink>
        <StyledLink to="/user/123">User Profile</StyledLink>
      </Nav>
    </HeaderContainer>
  );
}

export default Header;
```



3.2 Paste the code below in **Footer.jsx:**

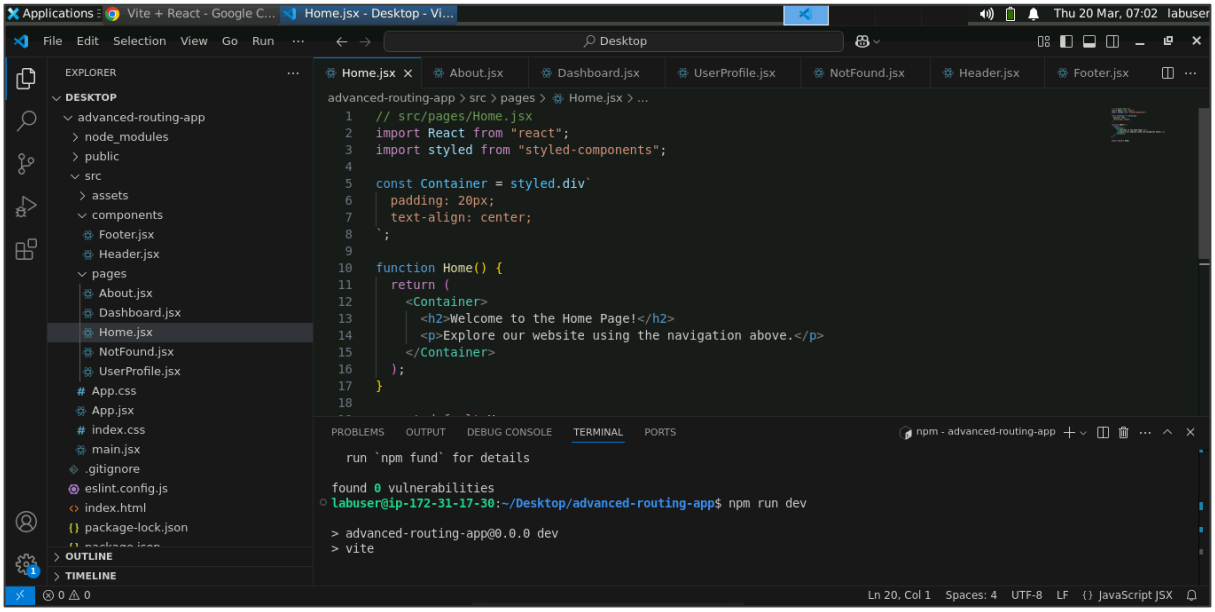
```
// src/components/Footer.jsx
import React from "react";
import styled from "styled-components";

const FooterContainer = styled.footer`
  background: #222;
  color: white;
  text-align: center;
  padding: 1rem;
  position: fixed;
  bottom: 0;
  width: 100%;
`;

const SocialLinks = styled.div`
  margin-bottom: 0.5rem;
`;

const StyledLink = styled.a`
  color: #fff;
  margin: 0 10px;
  text-decoration: none;
  &:hover {
    color: #f39c12;
  }
```

```
`;

function Footer() {
  return (
    <FooterContainer>
      <SocialLinks>
        <StyledLink href="https://twitter.com"
target="_blank">Twitter</StyledLink>
        <StyledLink href="https://facebook.com"
target="_blank">Facebook</StyledLink>
        <StyledLink href="https://linkedin.com"
target="_blank">LinkedIn</StyledLink>
      </SocialLinks>
      <p>&copy; 2025 Routing in React</p>
    </FooterContainer>
  );
}

export default Footer;
```



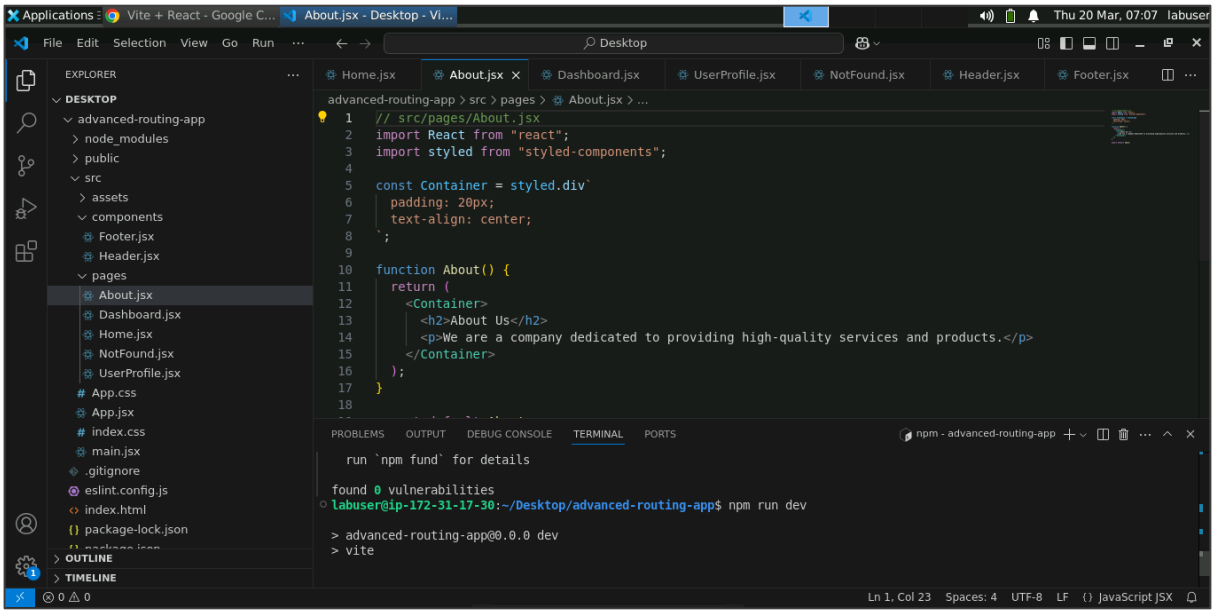
3.3 Paste the code below in **Home.jsx**:

```
// src/pages/Home.jsx
import React from "react";
import styled from "styled-components";

const Container = styled.div`
  padding: 20px;
  text-align: center;
`;

function Home() {
  return (
    <Container>
      <h2>Welcome to the Home Page!</h2>
      <p>Explore our website using the navigation above.</p>
    </Container>
  );
}

export default Home;
```



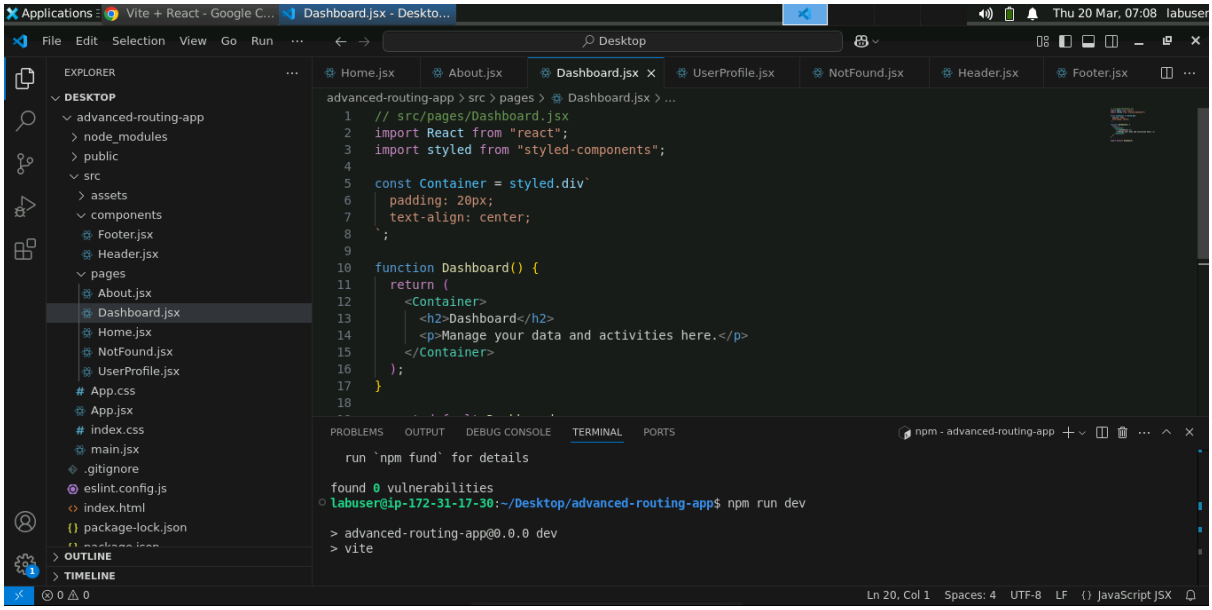
3.4 Paste the code below in **About.jsx**:

```
// src/pages/About.jsx
import React from "react";
import styled from "styled-components";

const Container = styled.div`
  padding: 20px;
  text-align: center;
`;

function About() {
  return (
    <Container>
      <h2>About Us</h2>
      <p>We are a company dedicated to providing high-quality services and
products.</p>
    </Container>
  );
}

export default About;
```



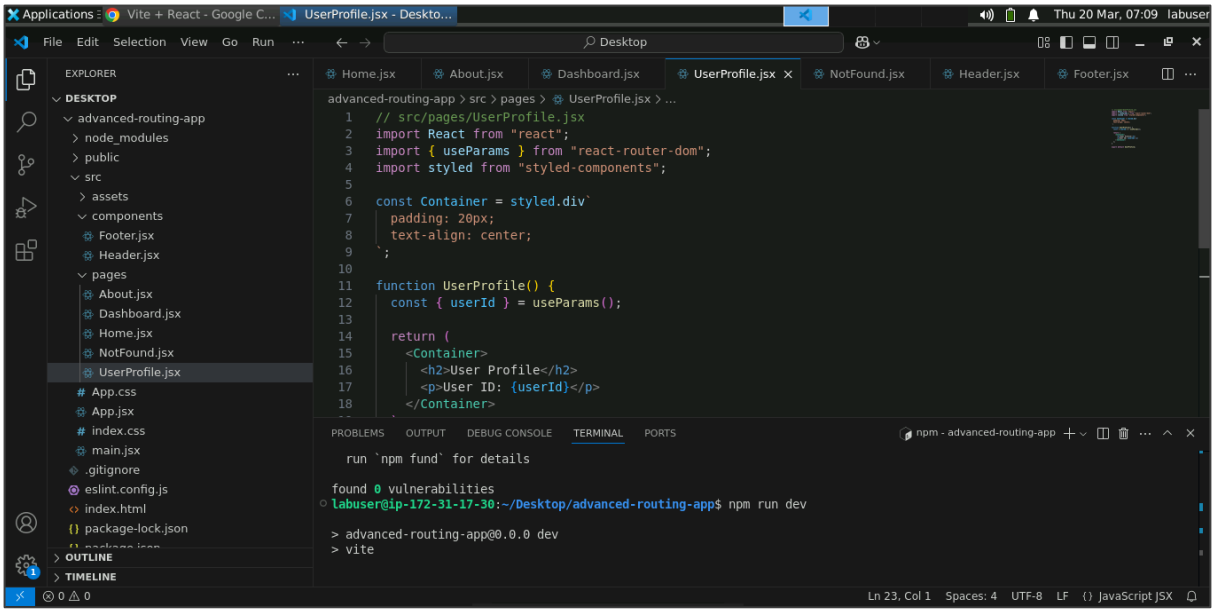
3.5 Paste the code below in **Dashboard.jsx**:

```
// src/pages/Dashboard.jsx
import React from "react";
import styled from "styled-components";

const Container = styled.div`
  padding: 20px;
  text-align: center;
`;

function Dashboard() {
  return (
    <Container>
      <h2>Dashboard</h2>
      <p>Manage your data and activities here.</p>
    </Container>
  );
}

export default Dashboard;
```



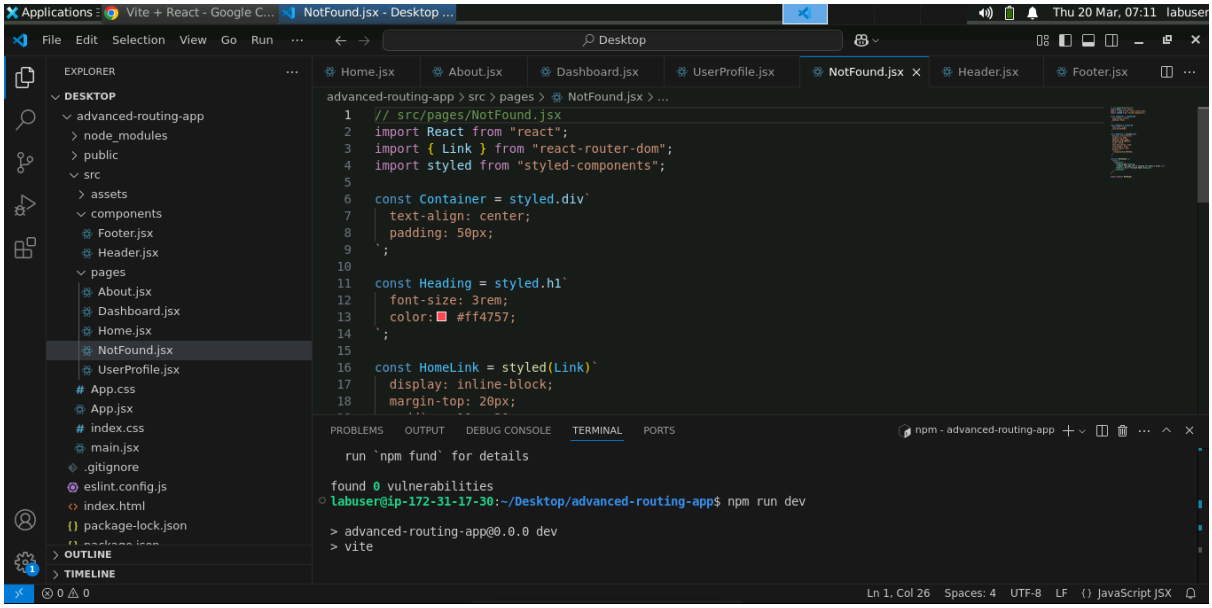
3.6 Paste the code below in **UserProfile.jsx**:

```
// src/pages/UserProfile.jsx
import React from "react";
import { useParams } from "react-router-dom";
import styled from "styled-components";

const Container = styled.div`
  padding: 20px;
  text-align: center;
`;

function UserProfile() {
  const { userId } = useParams();

  return (
    <Container>
      <h2>User Profile</h2>
      <p>User ID: {userId}</p>
    </Container>
  );
}

export default UserProfile;
```



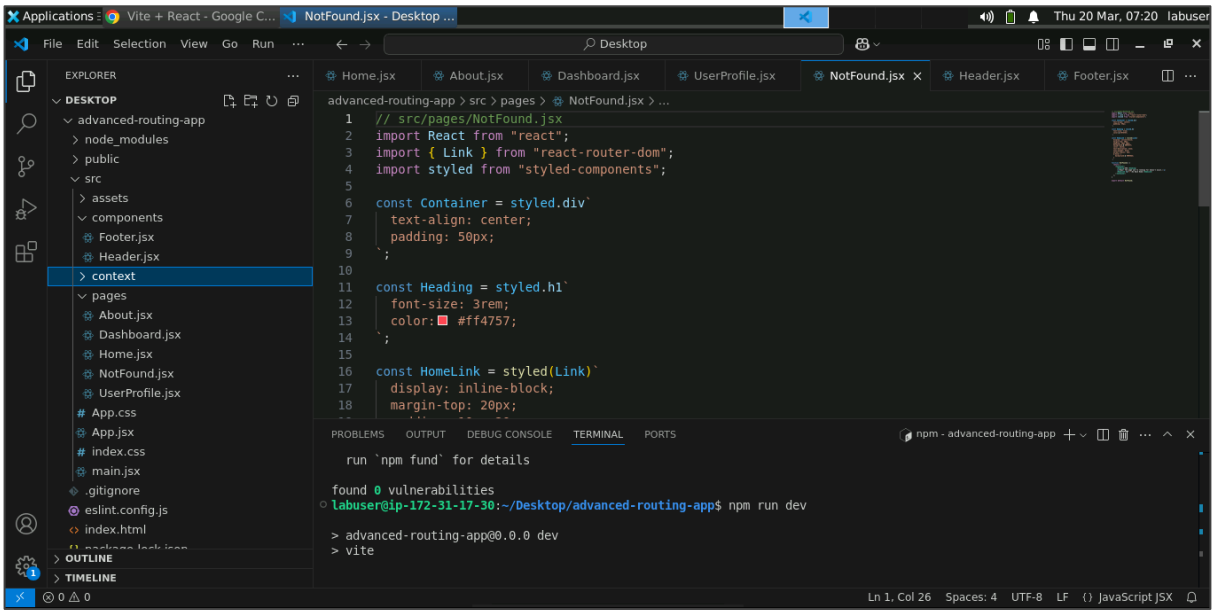
3.7 Paste the code below in **NotFound.jsx**:

```
// src/pages/NotFound.jsx
import React from "react";
import { Link } from "react-router-dom";
import styled from "styled-components";

const Container = styled.div`
  text-align: center;
  padding: 50px;
`;

const Heading = styled.h1`
  font-size: 3rem;
  color: #ff4757;
`;

const HomeLink = styled(Link)`
  display: inline-block;
  margin-top: 20px;
  padding: 10px 20px;
  background: #007bff;
  color: white;
  text-decoration: none;
  font-weight: bold;
  border-radius: 5px;
  &:hover {
    background: #0056b3;
  }
`;

function NotFound() {
  return (
    <Container>
      <Heading>404</Heading>
      <p>Oops! The page you're looking for doesn't exist.</p>
      <HomeLink to="/">Go Back Home</HomeLink>
    </Container>
  );
}

export default NotFound;
```

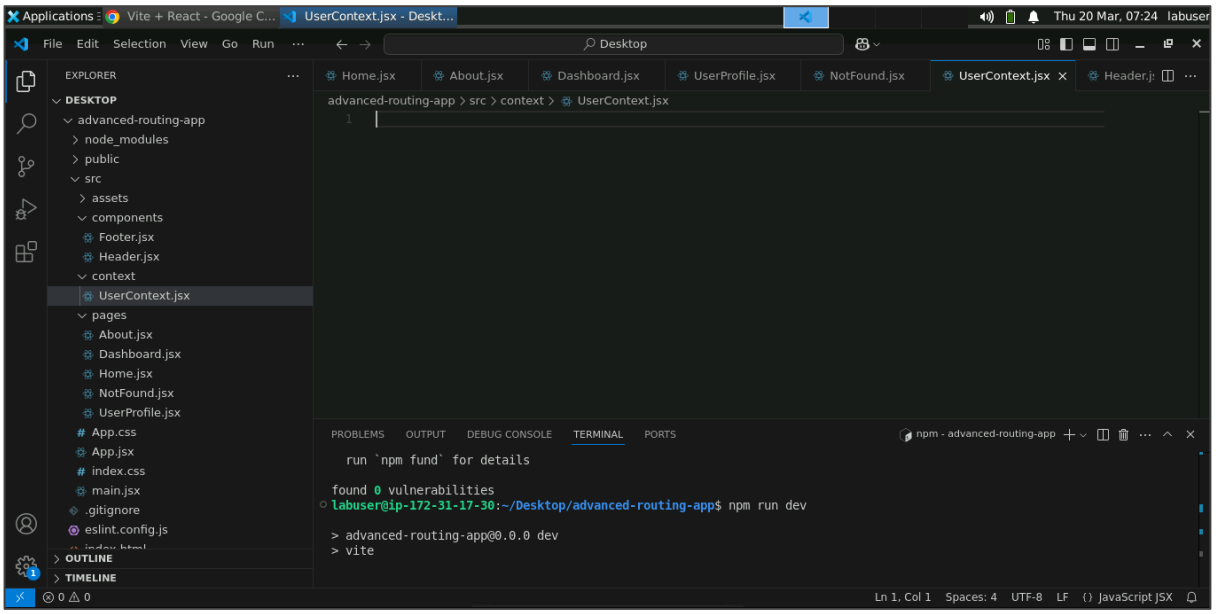


Step 4: Set up context for user management

4.1 Inside the **src** folder, create a new folder named **context**

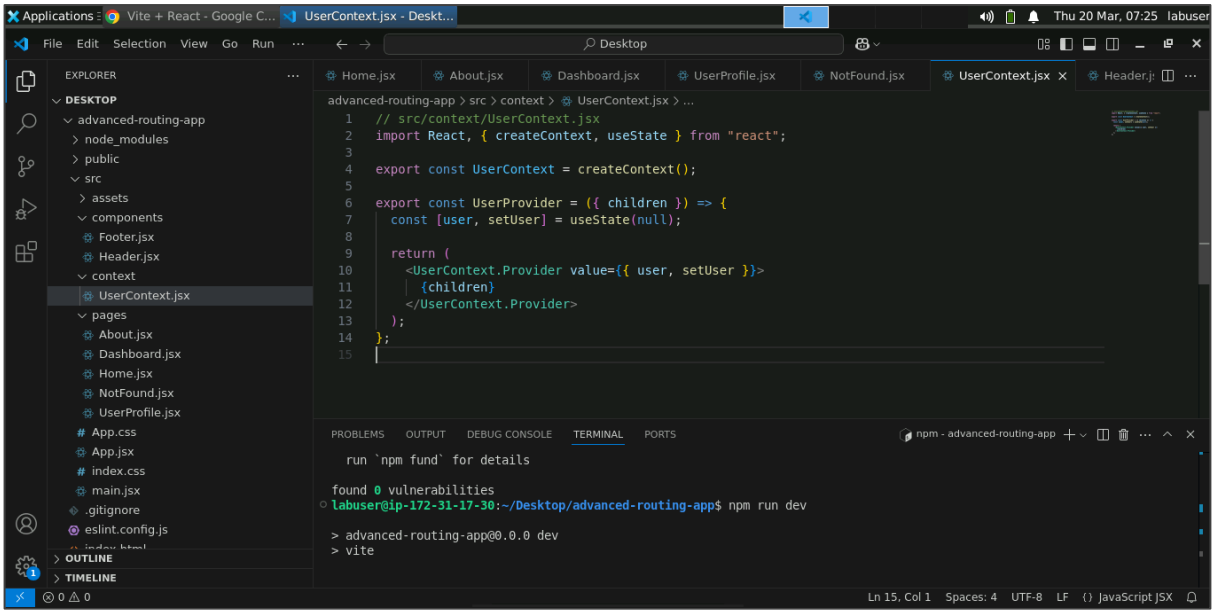


4.2 Inside the **context** folder, create a new file named **UserContext.jsx**



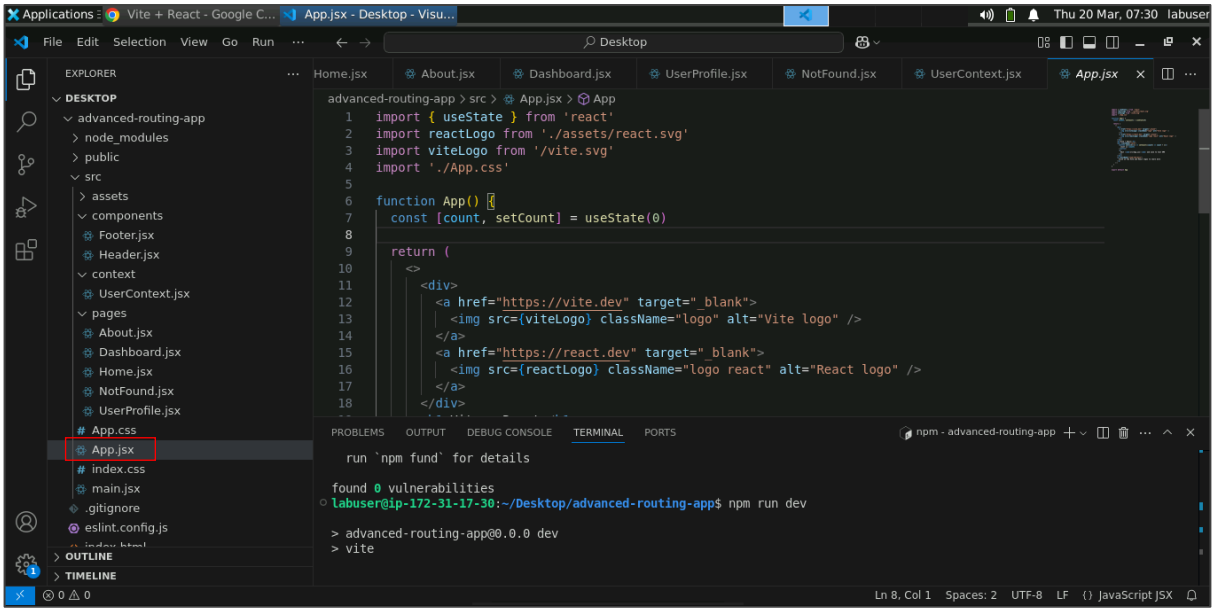
4.3 Paste the code below in **UserContext.jsx**:

```
// src/context/UserContext.jsx
import React, { createContext, useState } from "react";

export const UserContext = createContext();

export const UserProvider = ({ children }) => {
  const [user, setUser] = useState(null);

  return (
    <UserContext.Provider value={{ user, setUser }}>
      {children}
    </UserContext.Provider>
  );
};
```



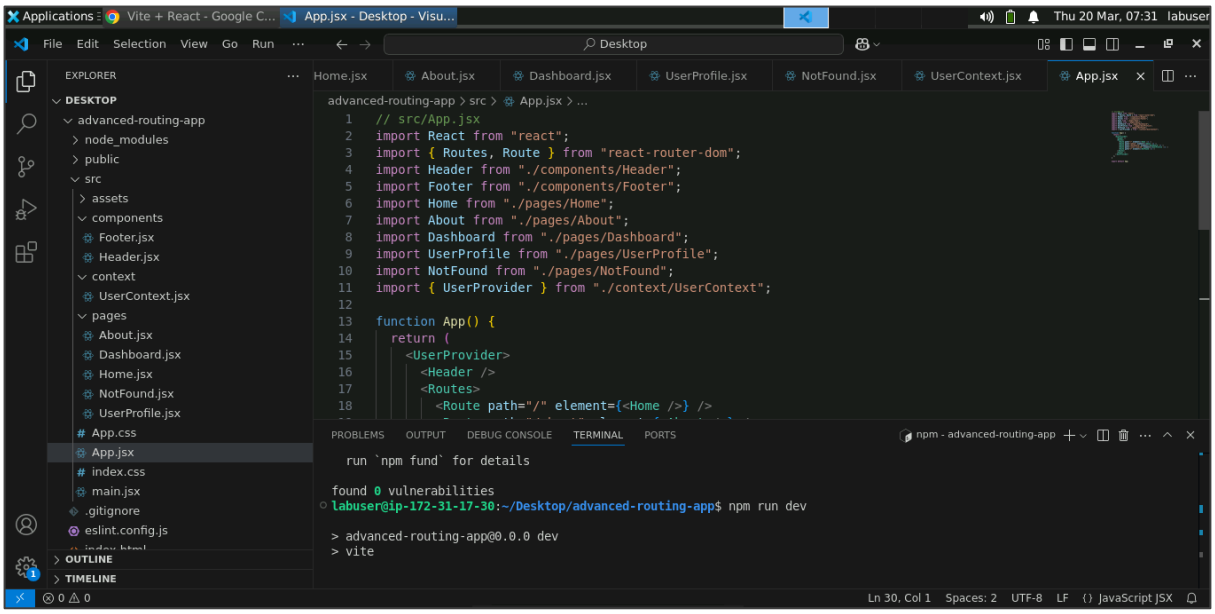
Step 5: Set up routing in App.jsx

5.1 Open the **App.jsx** file inside the **src** folder



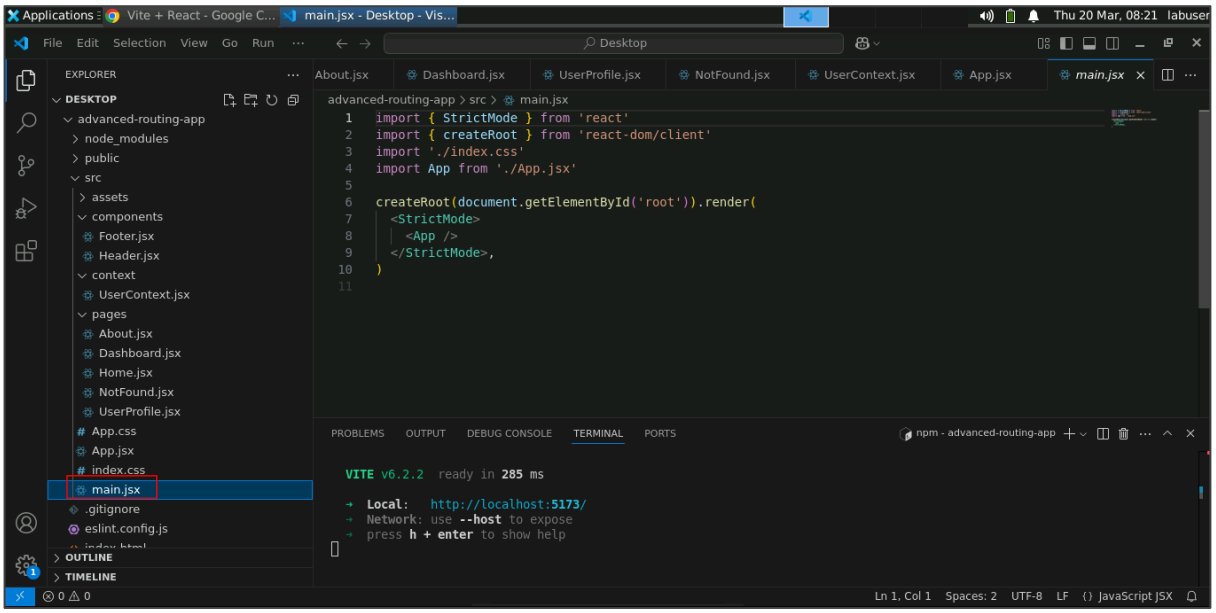
5.2 Paste the code below to include the new components in **App.jsx**:

```
// src/App.jsx
import React from "react";
import { Routes, Route } from "react-router-dom";
import Header from "./components/Header";
import Footer from "./components/Footer";
import Home from "./pages/Home";
import About from "./pages/About";
import Dashboard from "./pages/Dashboard";
import UserProfile from "./pages/UserProfile";
import NotFound from "./pages/NotFound";
import { UserProvider } from "./context/UserContext";

function App() {
  return (
    <UserProvider>
      <Header />
      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/about" element={<About />} />
        <Route path="/dashboard" element={<Dashboard />} />
        <Route path="/user/:userId" element={<UserProfile />} />
        <Route path="*" element={<NotFound />} />
      </Routes>
      <Footer />
    </UserProvider>
  );
}

export default App;
```



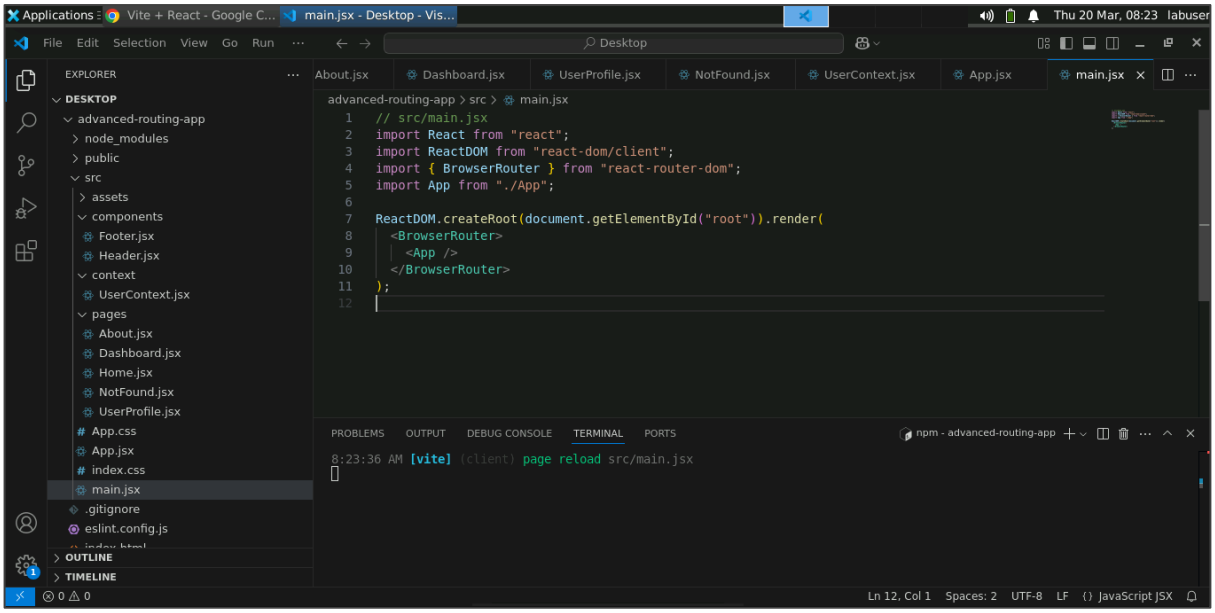
Step 6: Update main.jsx

6.1 Open the **main.jsx** file inside the **src** folder



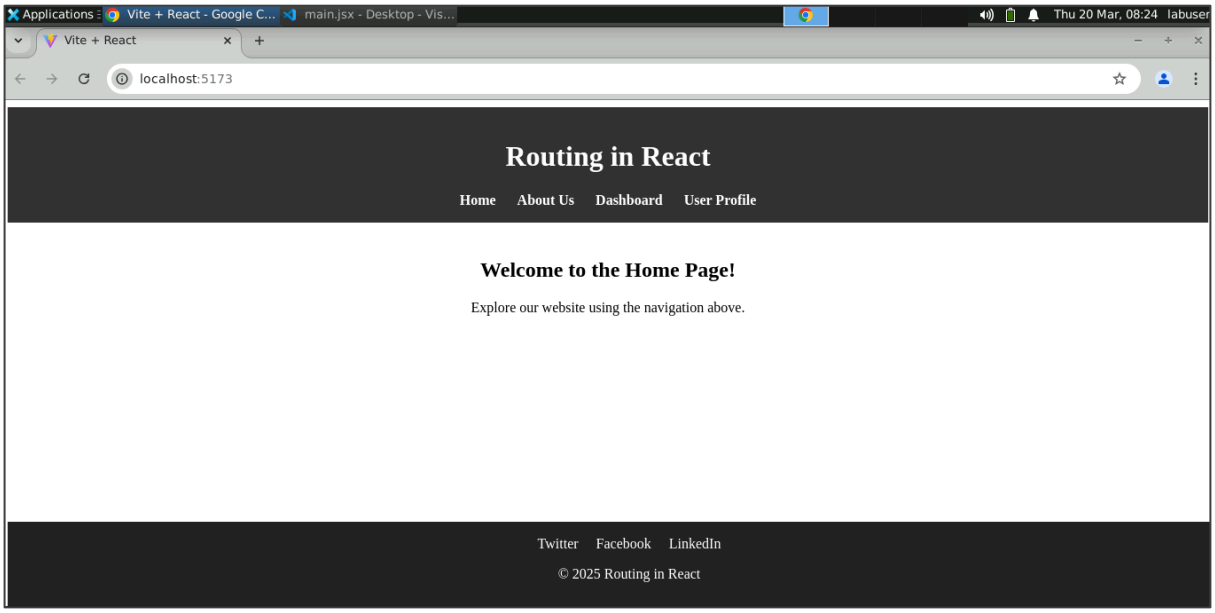
6.2 Paste the code below to include the new components in **main.jsx**:

```
// src/main.jsx
import React from "react";
import ReactDOM from "react-dom/client";
import { BrowserRouter } from "react-router-dom";
import App from "./App";

ReactDOM.createRoot(document.getElementById("root")).render(
  <BrowserRouter>
    <App />
  </BrowserRouter>
);
```

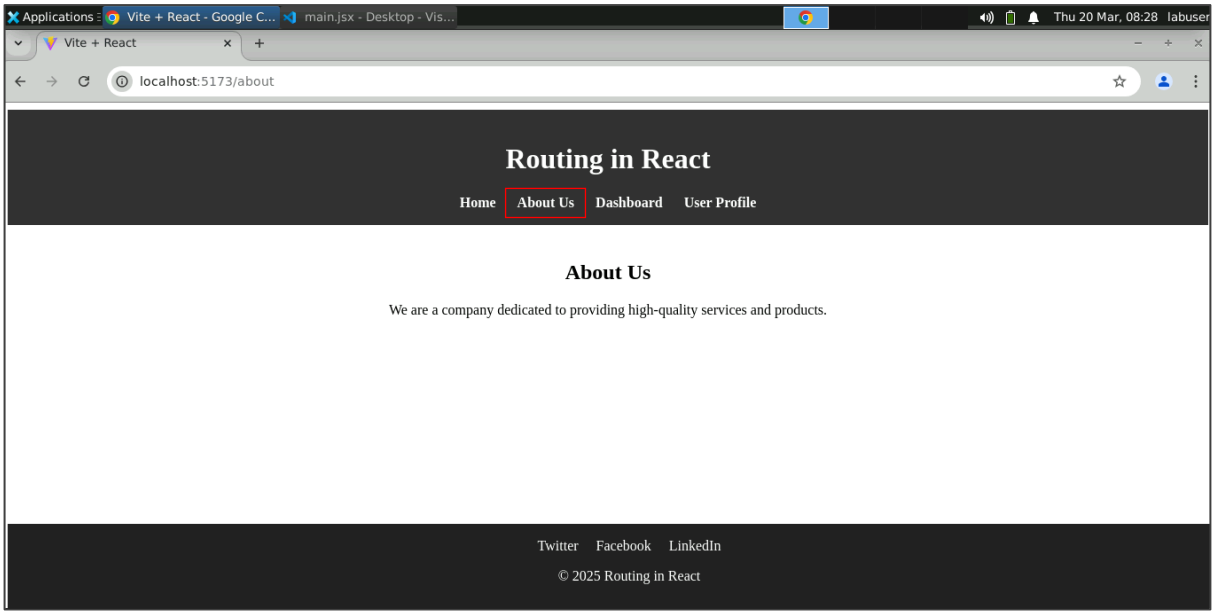


Step 7: Run and verify the application

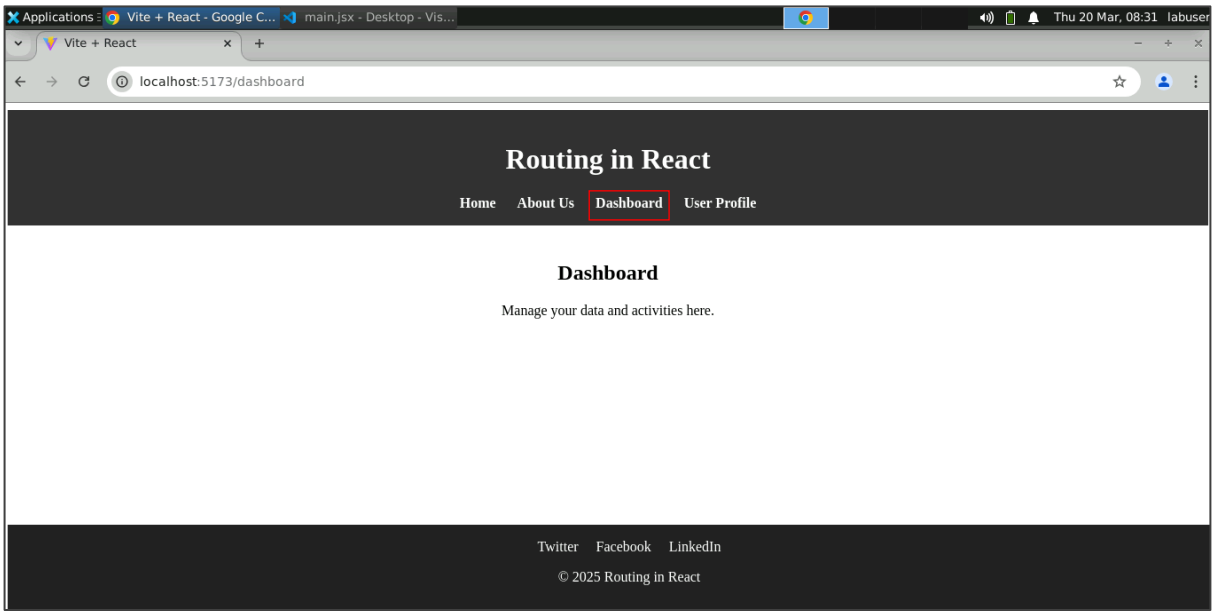
7.1 Navigate to the browser and refresh the webpage



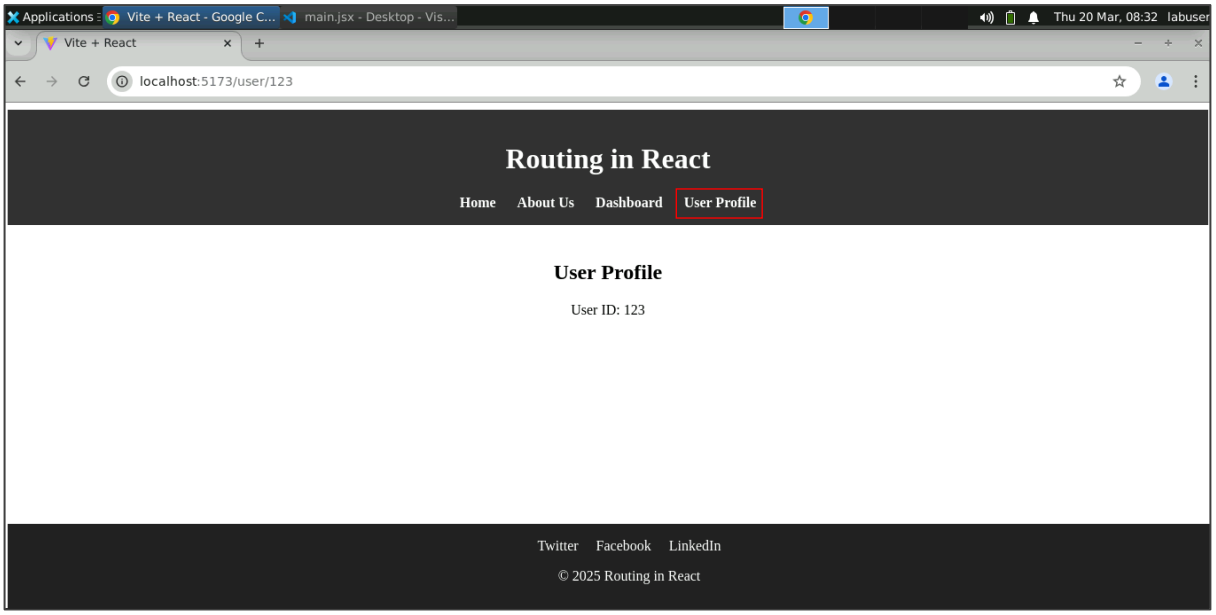
7.2 Click **About Us** to navigate to **/about** route



7.3 Click **Dashboard** to navigate to **/dashboard** route

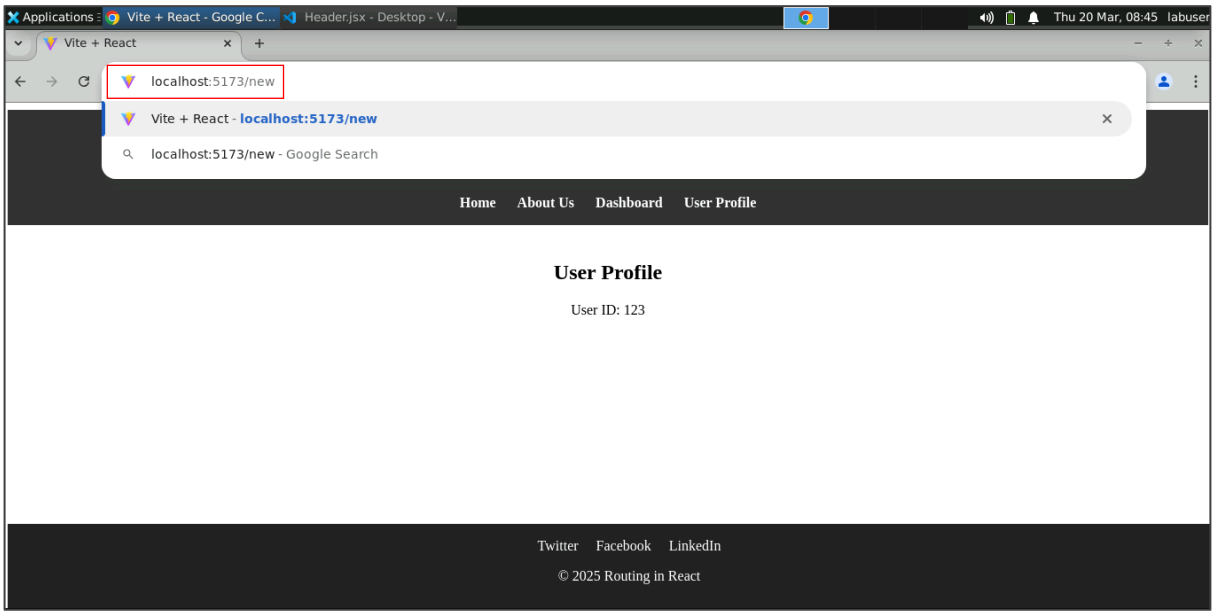


7.4 Click **User Profile** to navigate to **/user** route

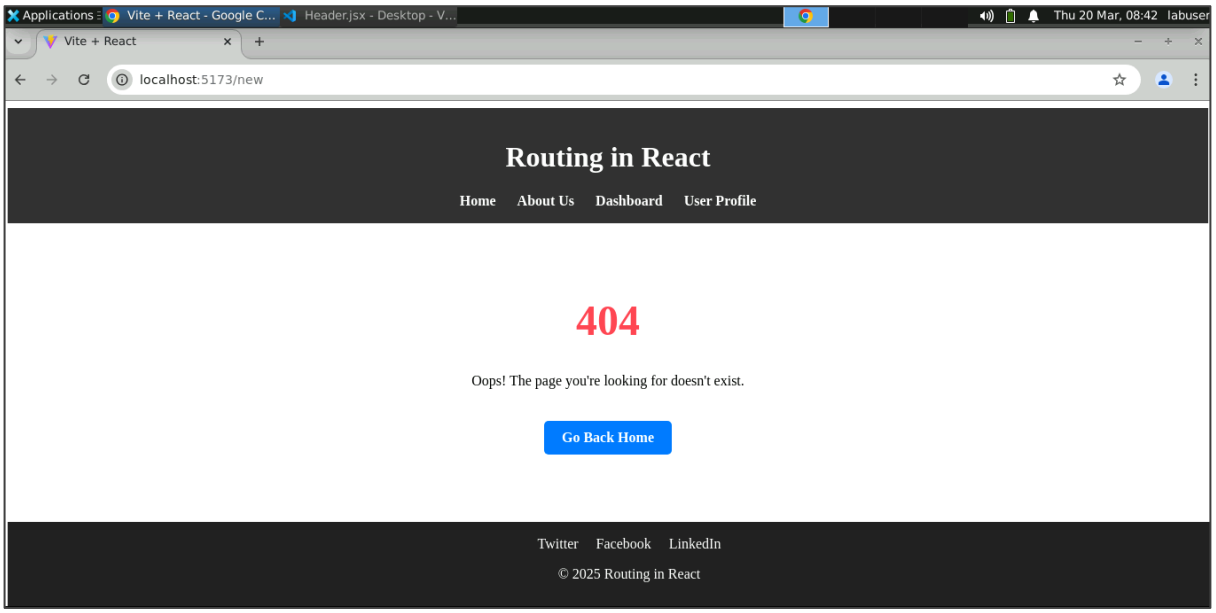


Note: This is a dynamic route that updates based on the **userId** provided in the URL.

7.5 Navigate to the URL search panel, enter any arbitrary word, and press **Enter**



The output for the page not found is as shown below:



By following the above steps, you have successfully built a React application with advanced routing using Vite. The app features structured navigation, dynamic routes, authentication-based access control, a custom 404 page, and smooth page transitions. This project provides a strong foundation for implementing scalable and user-friendly navigation in React applications.